

Bioinfo Lab 2

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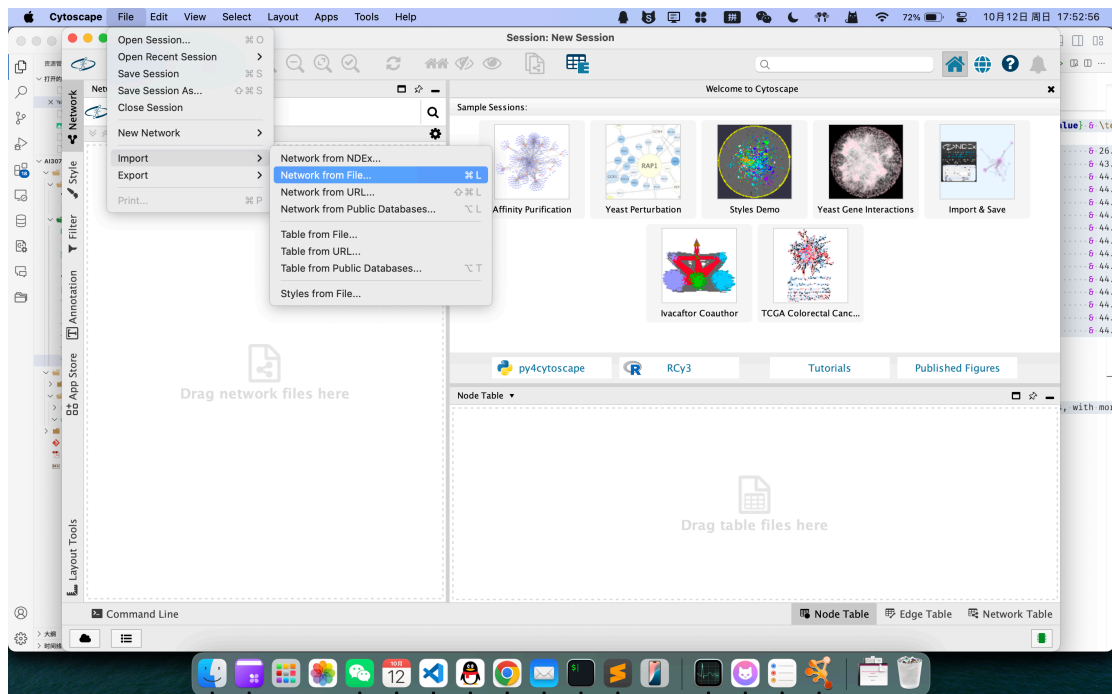


Figure 1

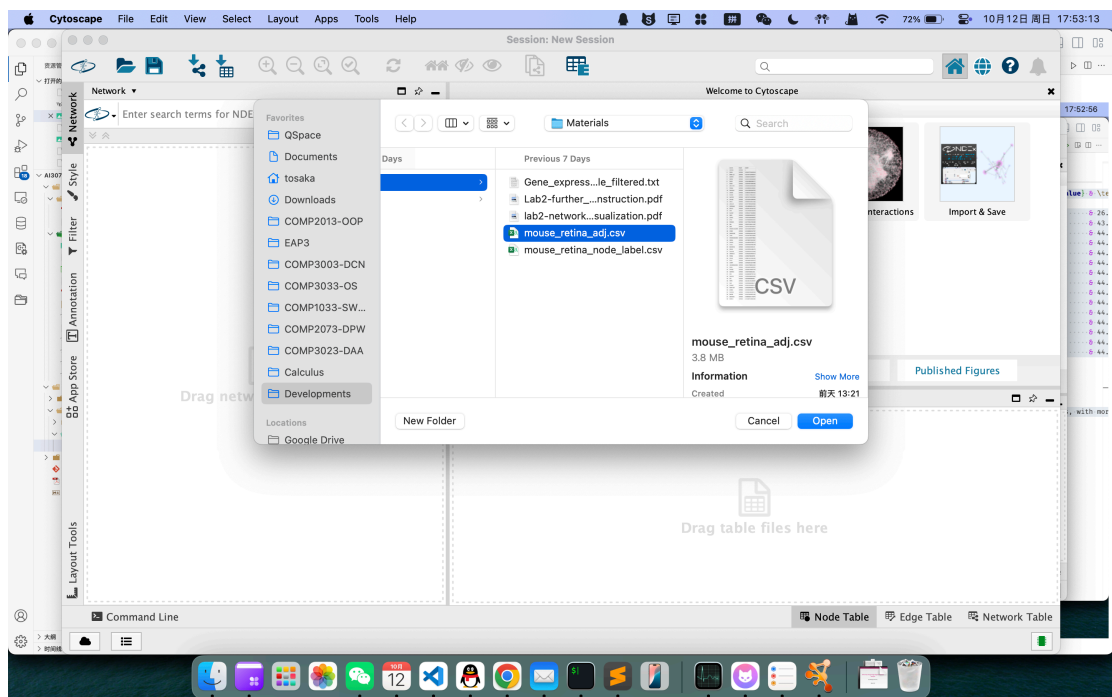


Figure 2

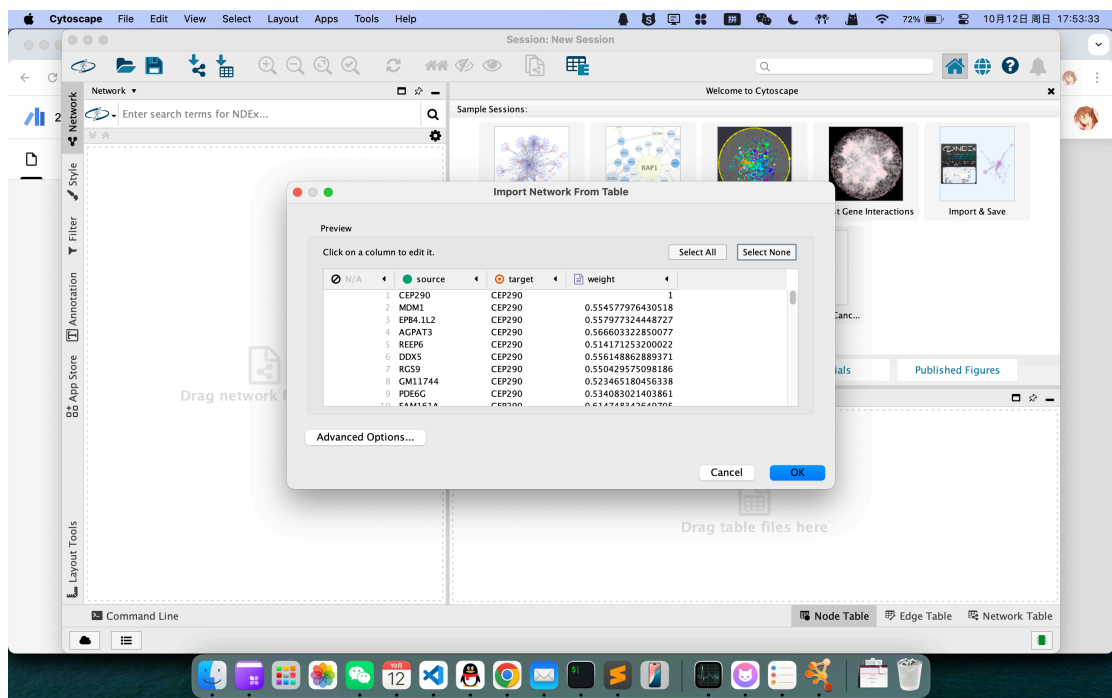


Figure 3

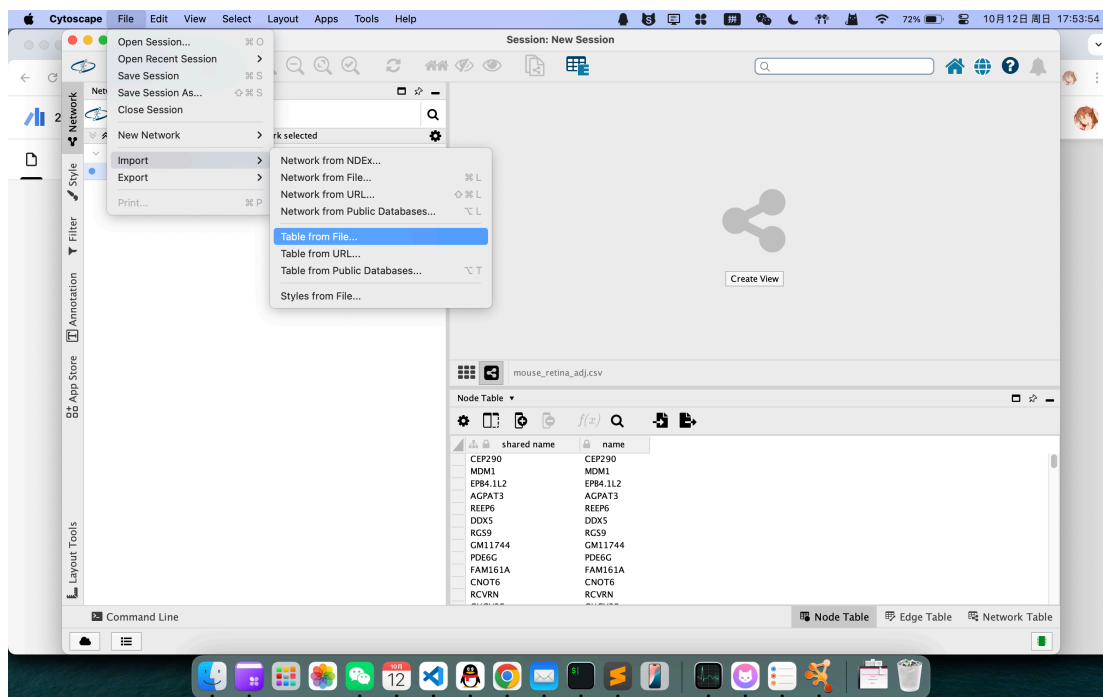


Figure 4

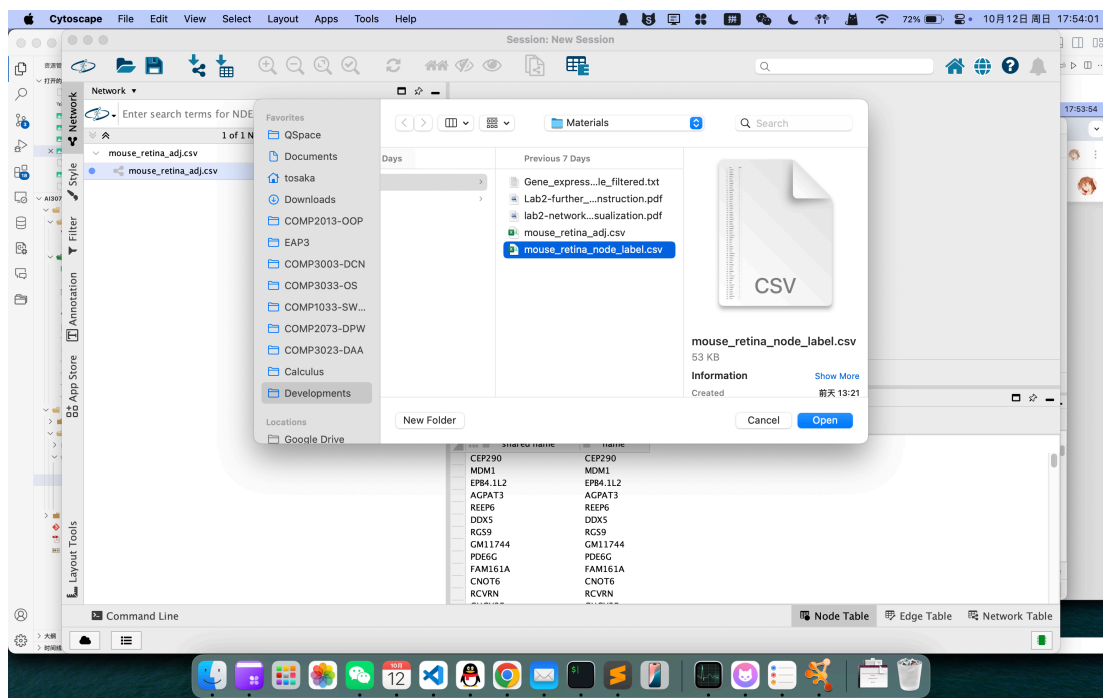


Figure 5

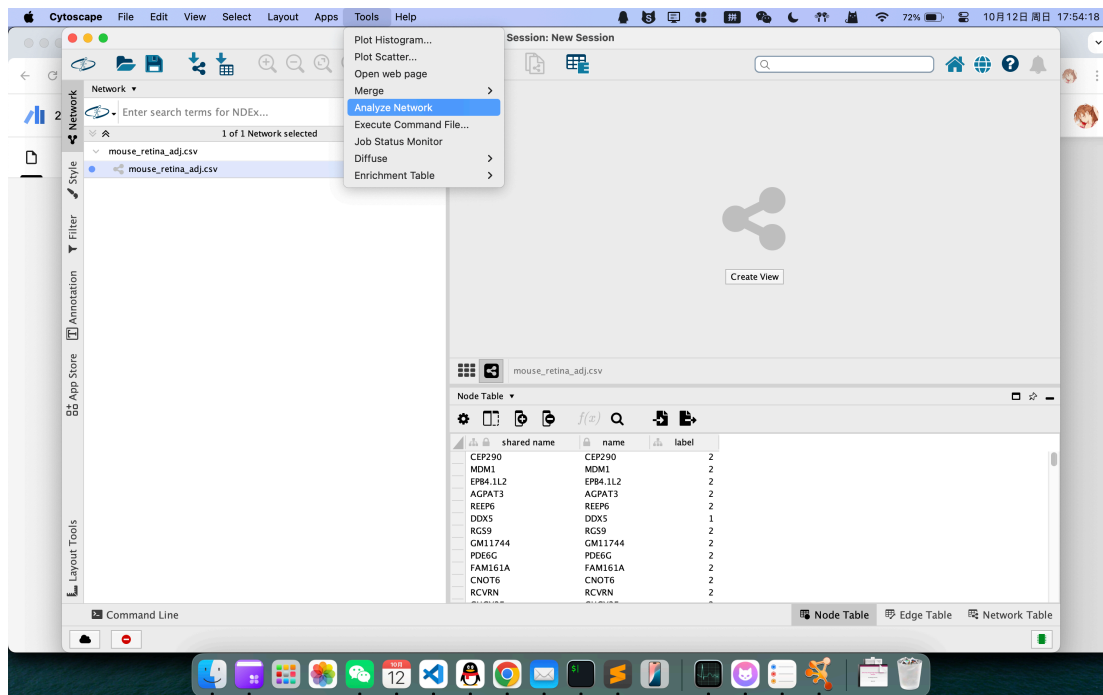


Figure 6

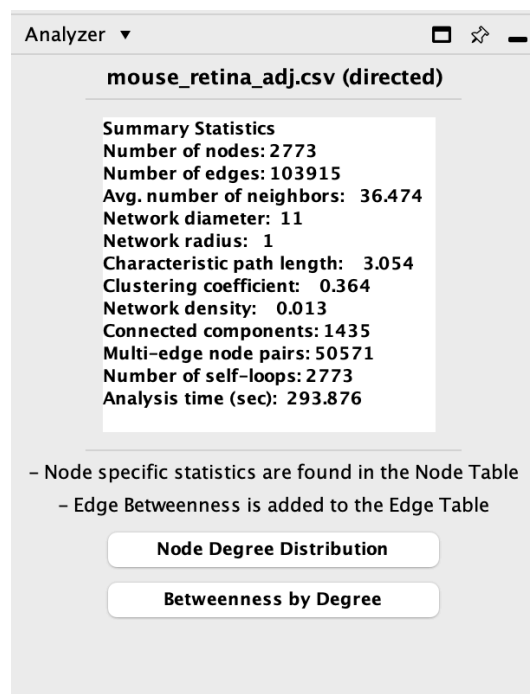


Figure 7

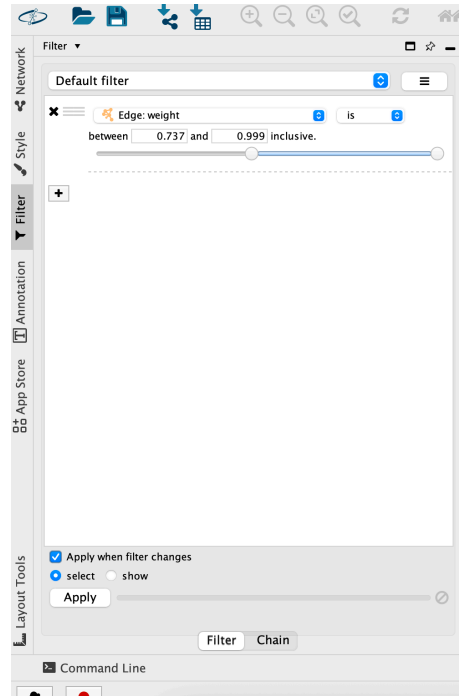


Figure 8

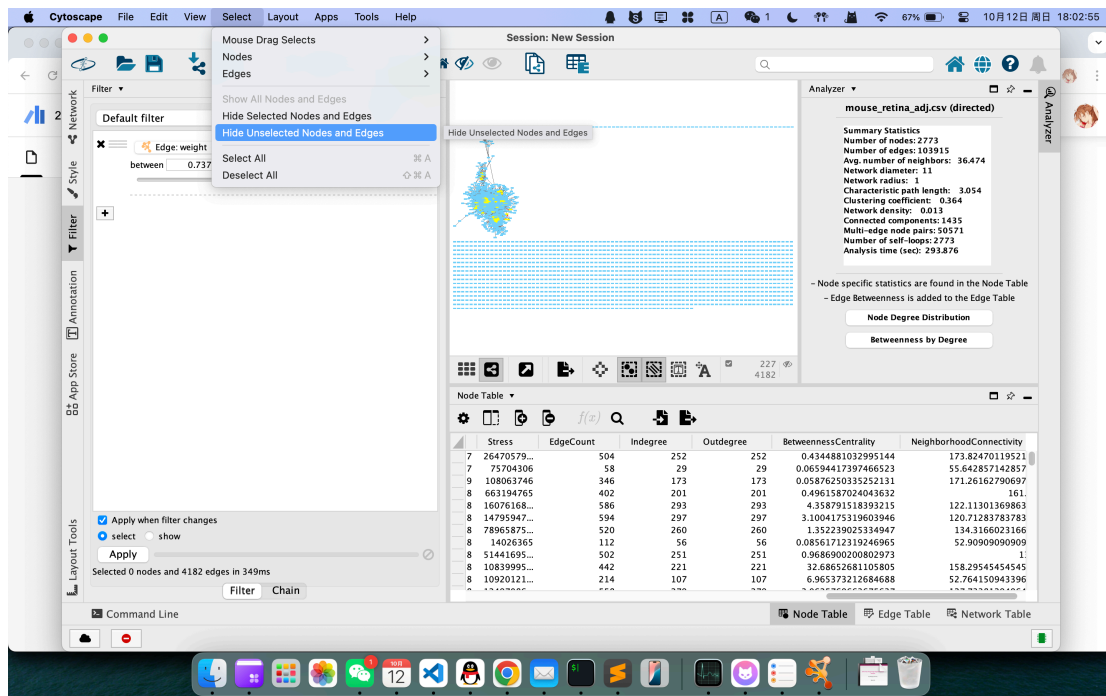


Figure 9

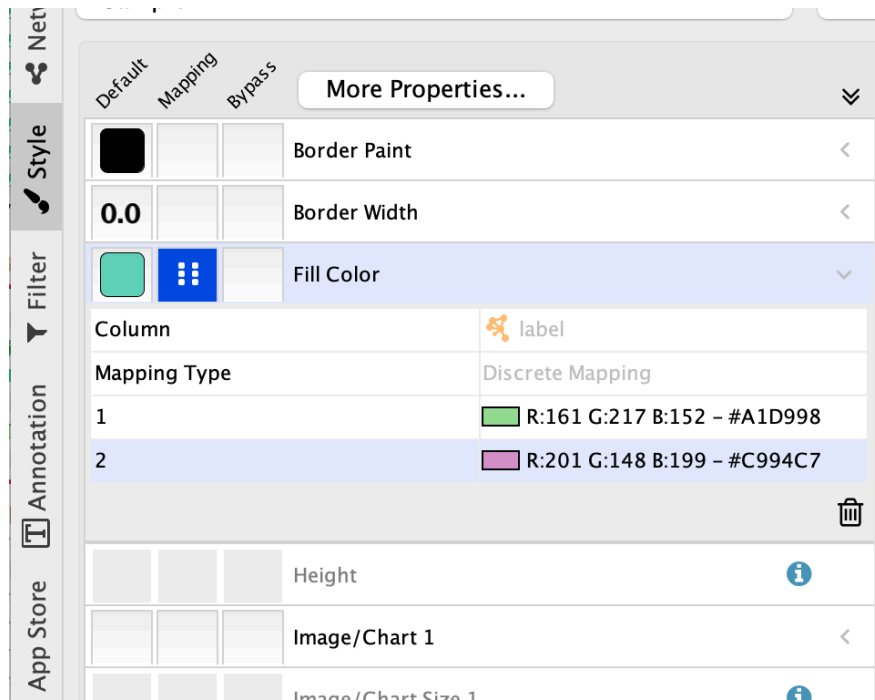


Figure 10

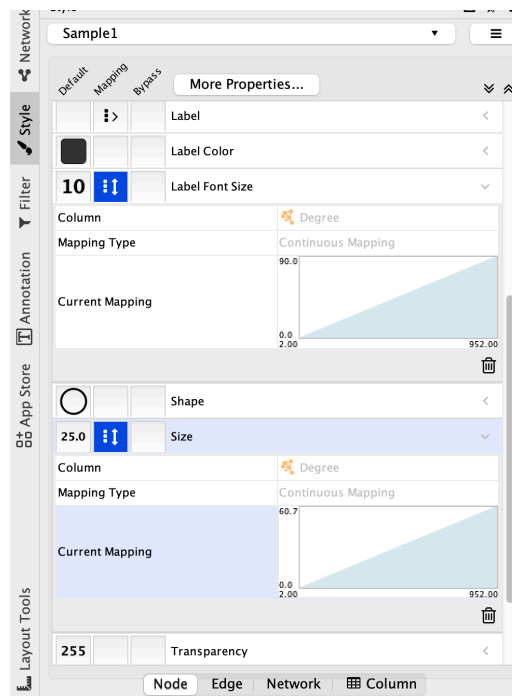


Figure 11

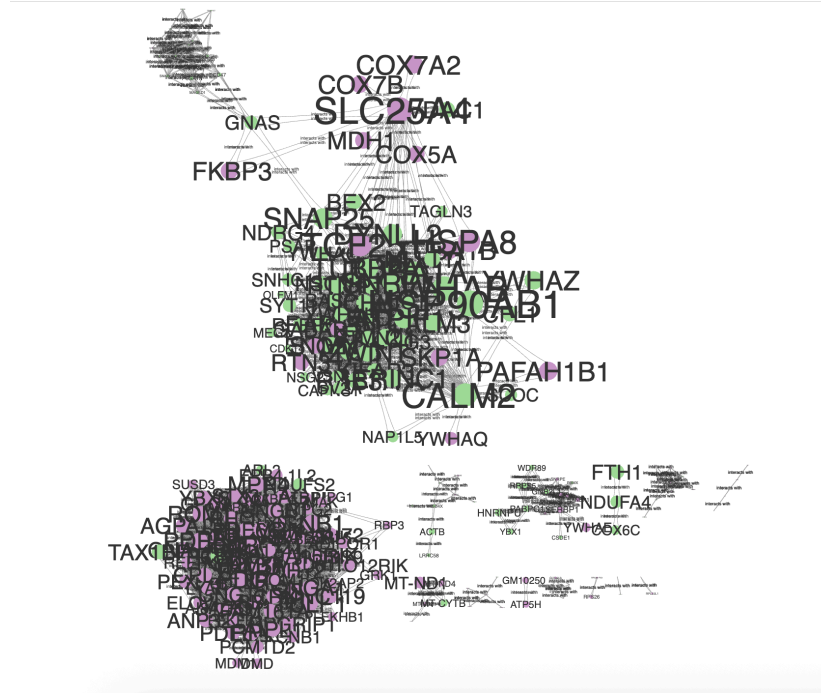


Figure 12

Key genes in the network map to well-defined functions on GeneCards: **CALM2** encodes calmodulin, a Ca^{2+} sensor regulating many enzymes and ion channels; **SNAP25** mediates synaptic vesicle exocytosis; **PAFAH1B1** (**LIS1**) controls neuronal migration via dynein–microtubule regulation; **FTH1** is the ferritin heavy chain for cellular iron storage; **SLC24A1** (**NCKX1**) is a K^{+} -dependent $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger essential for phototransduction; **GNAS** encodes a G-protein α -subunit driving cAMP signaling; and multiple mitochondrial energy genes—**COX5A**/**COX7A2** (complex IV), **SLC25A4**/**ANT1** (ADP/ATP carrier), **MDH1** (malate dehydrogenase)—underline enriched oxidative-phosphorylation and metabolic control in the hub set.